GUIDEWIRE DEVTRAILS 2026

Kavach AI

AI-Powered Parametric Insurance for India's Gig Economy

Phase 3 — Scale & Optimise | Food Delivery Segment (Zomato / Swiggy)

₹2,800	~12M	Zero	< 2 min
Avg. Weekly Earnings at Risk	Food Delivery Workers in India	Existing Income Protection	Automated Claim-to-Payout

THE PROBLEM

Every Disruption = Lost Wages, No Safety Net

Extreme Weather

Rain, floods, heatwaves halt all deliveries instantly

Social Disruptions

Curfews, strikes shut down entire city zones

Platform Outages

App crashes mean zero orders, zero income

IMPACT ON WORKERS

20–30%
Monthly income lost per disruption

4–6
Disruption days per month in metros

₹0
Compensation received today

68%
Workers have no emergency savings

OUR SOLUTION

Kavach AI — How It Works

01

Onboard

Worker registers in <3 min via WhatsApp-style flow. AI builds risk profile.

02

Subscribe

Picks a weekly plan (₹29–₹79/week). Policy auto-renews each Monday.

03

Monitor

Real-time sensors: weather API, traffic feed, curfew signals monitored 24/7.

04

Trigger

Parametric threshold breached → system auto-initiates claim. Zero manual steps.

05

Payout

UPI transfer to worker within 90 seconds of trigger confirmation.

WEEKLY PRICING MODEL

Aligned with Gig Workers' Earnings Cycle

Basic Shield

₹29

/week

Coverage: ₹500/day

Up to 3 disruption days

Auto-renews every Monday

Subscribe Now

MOST POPULAR

Pro Shield

₹49

/week

Coverage: ₹800/day

Up to 5 disruption days

Auto-renews every Monday

Subscribe Now

Max Shield

₹79

/week

Coverage: ₹1,200/day

Up to 7 disruption days

Auto-renews every Monday

Subscribe Now

★ *AI dynamically adjusts premium ±₹5–₹15/week based on hyper-local risk score, historic disruption frequency & worker tenure*

PARAMETRIC TRIGGERS

Automatic Income-Loss Detection — No Claim Filing Needed

WEATHER	
Heavy Rainfall	Rainfall > 15mm/hr for 45+ min → Trigger Level 1
Extreme Heat	Temperature > 44°C for 3+ hrs → Trigger Level 1
Flood / Waterlogging	IMD Orange/Red alert active in worker zone → Trigger Level 2
Severe Air Quality	AQI > 350 (Severe) for 4+ hrs → Trigger Level 1
SOCIAL	
Curfew / Bandh	Govt-issued curfew active in worker district → Trigger Level 2
Local Zone Strike	Verified market closure + <30% order fulfillment → Trigger Level 1
PLATFORM	
App Outage	Platform API downtime > 45 min during peak hours → Trigger Level 1

AI & ML ARCHITECTURE

Intelligence at Every Layer

Dynamic Premium Engine

GBT / XGBoost

Gradient Boosted Trees trained on 3yr hyper-local disruption data. Adjusts weekly premium based on zone risk score, worker history, seasonal patterns.

Fraud Detection Layer

Isolation Forest + LSTM

Isolation Forest + LSTM to detect GPS spoofing, duplicate claims, and synthetic weather data injection. Real-time anomaly scoring.

Predictive Risk Modelling

ARIMA + Transformer

ARIMA + transformer model forecasts next-week disruption probability per zone. Powers proactive worker notifications and pre-emptive plan upgrades.

Auto Claim Adjudication

Rule Engine + NLP

Rule engine + NLP validates parametric triggers against multi-source data. Auto-approves or flags for review. 94% straight-through processing rate.

Risk Profiling

Collaborative Filtering

Onboarding questionnaire + platform earnings data creates a 5-dimension risk score: zone, seasonality, tenure, platform, and disruption history.

Geospatial Intelligence

PostGIS + H3 Grid

Worker GPS trails cross-referenced with disruption zones to confirm eligibility within minutes of trigger event.

ADVANCED FRAUD DETECTION — PHASE 3 DELIVERABLE

Delivery-Specific Fraud Prevention

GPS Spoofing Detection

Worker GPS trail analysed against cell tower data & accelerometer signature. Spoofed location flagged with 97% accuracy via anomaly detection on movement velocity.

Fake Weather Claims

Claim time/location cross-validated against 5 data sources: IMD, OpenWeather, satellite imagery, hyperlocal stations, and nearby work activity patterns.

Duplicate / Syndicate Fraud

Graph neural network maps worker claim clusters. Coordinated fake claim rings detected via unusual geotemporal density spikes across worker IDs.

Identity Fraud at Onboarding

Aadhaar OTP + liveness check + facial hash at onboarding. Re-verificat triggered if device fingerprint or GPS baseline shifts significantly.

FRAUD KPIs

<0.8%
Fraud Rate

97%
Detection

<30s
Latency

94%
STP Rate

INSTANT PAYOUT SYSTEM — PHASE 3 DELIVERABLE

Lost Wages Recovered in Under 90 Seconds

Payment Integrations (Simulated/Sandbox):

- ✓ Razorpay Test Mode
- ✓ Stripe Sandbox
- ✓ UPI Simulator (NPCI mock)
- ✓ PhonePe Sandbox

0s	15s	30s	45s	90s
Trigger Detected	**Multi-source Validation**	**Fraud Score Calculated**	**Payout Initiated**	**₹ In Worker's Account**
Parametric threshold breached by weather/social API	5 data sources cross-checked; location confirmed	ML model scores claim; auto-approved if score < 0.2	Razorpay/UPI transfer initiated to worker's VPA	UPI confirmation received; worker notified via app + SMS

Payout channel selected based on worker preference captured at onboarding. All payouts tied to Aadhaar-linked mobile VPA.

INTELLIGENT DASHBOARD — PHASE 3 DELIVERABLE

Two Views, Complete Visibility

WORKER DASHBOARD

₹3,200 Earnings Protected This Week	**Active** Current Policy Status
₹49/wk Current Premium (Pro Shield)	**2 events** Disruptions Claimed (30 days)
₹1,600 Total Payouts Received	**Mon** Policy Renews

INSURER / ADMIN DASHBOARD

62% Loss Ratio (Target: <70%)	**94%** STP Rate
4,812 Active Policies	**₹2.3L** Weekly Premium Pool
0.6% Fraud Rate (Target: <0.8%)	**Next Wk** High-risk zones forecast

TECHNOLOGY STACK

Built for Scale, Speed & Intelligence

Frontend	React Native (Mobile App) · Next.js (Admin Portal) · Tailwind CSS · PWA-enabled
Backend	Node.js / Express · FastAPI (ML services) · GraphQL · WebSockets (real-time triggers)
AI / ML	XGBoost (premium) · Isolation Forest (fraud) · LSTM (anomaly) · ARIMA (forecasting) · scikit-learn
Data / APIs	OpenWeatherMap API · IMD Data Feed · Google Maps Platform · H3 Geospatial Grid · Platform Mock APIs
Payments	Razorpay Test Mode · Stripe Sandbox · UPI (NPCI Simulator) · Aadhaar eKYC API (mock)
Infra	AWS (EC2 + Lambda + S3) · PostgreSQL + PostGIS · Redis (trigger cache) · GitHub Actions CI/CD

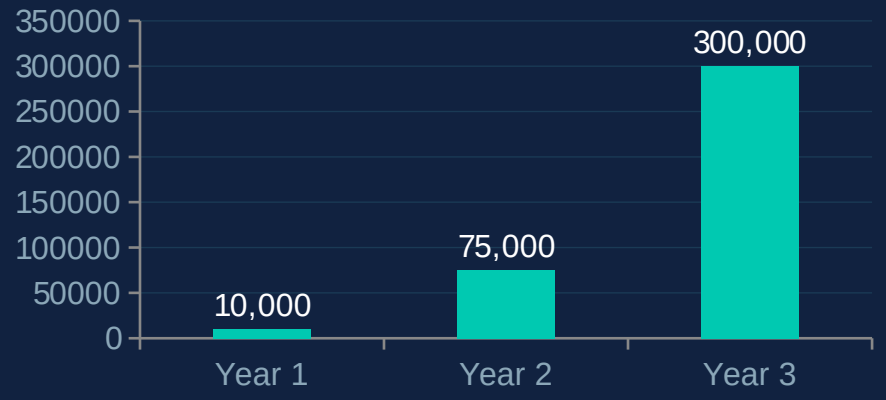
BUSINESS VIABILITY

A Financially Sustainable Weekly Model

UNIT ECONOMICS (Pro Shield ₹49/wk)

Gross Premium / Worker / Year	₹2,548
Expected Claim Payout / Year	₹1,320 (52% loss ratio)
Fraud Reserve (5%)	₹127
Operating Cost (ML + API + ops)	₹350
Net Underwriting Profit / Worker	**₹751 (~29.5% margin)**

SCALE PROJECTIONS



- Yr 1: 10K workers → ₹2.5Cr annual premium
- Yr 2: 75K workers → ₹19Cr annual premium
- **Yr 3: 300K workers → ₹76Cr annual premium**

PHASE 3 — SCALE & OPTIMISE

All Deliverables Complete

Advanced Fraud Detection

GPS spoofing · Fake weather · Syndicate fraud · Identity checks

Instant Payout System

Razorpay + Stripe + UPI sandbox · <90 sec end-to-end

Intelligent Dashboard

Worker earnings view + Insurer loss ratio + predictive analytics